

Charli D. Minsavage-Davis

Doctoral Research Assistant

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Department of Biology

Georgetown University

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Education

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|------|---|--|
| 2027 | Ph.D. Georgetown University
Advisor: Matthew Hamilton | Focus: Population Genetics and Bioinformatics
Committee: Shweta Bansal, Rodney Dyer, and Sivan Leviyang |
| 2021 | M.S. Ohio State University
Advisor: G. Matthew Davies | Focus: Quantitative Ecology
Committee: Stephen Matthews, Roger Williams |
| 2018 | B.S. Ohio State University
Academic Advisor: Brian Slater | Research Advisor: G. Matthew Davies |

External Grants and Fellowships

- 2025 **Metropolitan Washington Chapter Scholar**, ARCS Foundation, Inc. (\$15,000)
2025 **Bill Dahl Graduate Student Research Award**, Botanical Society of America (\$1,500)
2024 **Research Award**, Washington Biologists' Field Club (\$3,000)

Internal Grants and Fellowships

- 2023 **GradGov – GSAS Research Grant**, Georgetown University (\$350)
2019 **Patrick S. Osmer Graduate Research Fellowship**, Ohio State University (\$25,800)
2018 **Summer Research Opportunities Program (SROP)**, Big Ten Academic Alliance (\$4,000)

Publications

**First or shared first author*

6. Hamilton, M.B.; **Minsavage-Davis, C.D.**; Kottler, E.J.; Oyler, E.; Peterson, A.; Johnson, B.A.; Gedan, K. 2025. Clonal genetic architecture and landscape genetics of the foundation plant *Spartina patens* in shrinking high salt marsh habitats. **In preparation.**
5. **Minsavage-Davis, C.D.***; Davies, G.M.; et al. 2025. Fuel management and traditional managed burning reduce potential landscape-scale rates of spread and fireline intensities in northern-European heathlands. **In preparation.**
4. **Minsavage-Davis, C.D.***; Davies, G.M.; Haugum, S.V.; Velle, L.G.; Vandvik, V. 2024. Development and evaluation of generalized fuel models for predicting fire behaviour in northern European heathlands. *iForest*, 17, 109-119. <https://doi.org/10.3832/ifer4394-017>
3. Rippel, T.M.*; **Minsavage-Davis, C.D.***; Shirey, V.; Wimp, G.M. 2023. Simple machine learning with aerial imagery reveals severe loss of a salt marsh foundation species. *Estuaries and Coasts*, 46, 1110-1122. <https://doi.org/10.1007/s12237-023-01192-z>
2. **Minsavage-Davis, C.D.***; Garthwaite, I.J.; Fisher, M.D.; Leigh, A.; Ramstack Hobbs, J.M.; Claeson, S.M.; Wimp, G.M.; LeRoy, C.J. 2023. Spatial Habitat Structure Assembles Willow-Dependent Communities across the Primary Successional Watersheds of Mount St. Helens, USA. *Forests*, 14, 322. <https://doi.org/10.3390/f14020322>
1. **Minsavage-Davis, C.D.***; Davies, G.M. 2022. Evaluating the Performance of Fire Rate of Spread Models in Northern-European *Calluna vulgaris* Heathlands. *Fire*, 5, 46. <https://doi.org/10.3390/fire5020046>

Select Research presentations

**Presenting author*

- 2025 **Simple Machine Learning with Aerial Imagery Reveals Severe Loss of a Salt Marsh Foundation Species**, ESA Mid-Atlantic Region Meeting, Contributed Oral Session
2025 **Ecological genetics of Mid-Atlantic populations of *Spartina patens***, ESA Mid-Atlantic Region Meeting, Poster
2024 **Assessing the impacts of vertical forest structure on a common insect pest***, Spotlight on National Park Resources, Contributed Oral Session

- 2023 **Assessing the impacts of vertical forest structure on a common insect pest***, EntSoc Annual Meeting, Contributed Oral Session
- 2023 **How does local adaptation vary with diet breadth? A meta-analysis of local adaptation in specialist versus generalist herbivorous insects**, EntSoc Annual Meeting, Contributed Oral Session
- 2023 **Dietary parental effects in a generalist herbivore**, EntSoc Annual Meeting, Contributed Oral Session
- 2023 **Understanding the scope of diet breadth in a generalist herbivore**, EntSoc Annual Meeting, Contributed Oral Session
- 2023 **Spatial Habitat Structure Assembles Willow-Dependent Communities across the Primary Successional Watersheds of Mount St. Helens, USA***, ESA General Meeting, Contributed Oral Session
- 2023 **Clonal structure and landscape genetics of Long Island, NY populations of the salt marsh plant *Spartina patens****, ESA Mid-Atlantic Region Meeting, Poster
- 2022 **The grass is always greener: remote sensing and machine learning reveal severe and enigmatic loss of a salt marsh foundation species***, ESA General Meeting, Poster
- 2020 **Analysis of Spatial, Age, and Disturbance Patterns of a Forested Plot in Fairfield County Ohio***, City of Columbus Metro Parks, Professional Talk
- 2020 **Impact of Fuel Management Strategies on Potential Fire Behavior in the Heathlands and Moorlands of North-West Europe***, SER Midwest Chapter Meeting, Contributed Oral Session
- 2019 **A Physical and Chemical Stratigraphy of Peat Deposits in Flatiron Bog, northeast Ohio***, SER Midwest Chapter Meeting, Poster

Teaching experience

- 2024 AU **Genetics Teaching Assistant**, Georgetown University
- 2024 SP **Evolutionary Processes Teaching Assistant**, Georgetown University
- 2022 – 23 AU **Experimental Design Teaching Assistant**, Georgetown University
- 2021 AU **Instructor of Record – General Environmental Science I**, Northern Virginia Community College
- 2021 SP **Lab Section Leader for Introduction to Spatial Information (GIS)**, Ohio State University
- 2020 SU **Field Lecturer for Biology and Identification of Woody Forest Plants**, Ohio State University

Scientific Peer Review

- 2024 **Landscape Ecology**, Springer
- 2024 **International Journal of Wildland Fire**, CSIRO
- 2023 **Life & Environment**, PeerJ

Service

- 2023 – **Student Ambassador Program**, Georgetown University
- 2022 – 23 **Contract Bargaining Team Member**, Georgetown Alliance of Graduate Employees
- 2022 **Lead Organizer for Solidarity, Education, and Activism**, Georgetown Alliance of Graduate Employees
- 2019 – 21 **Professional Mentor for Underrepresented Students**, Ohio State University

Student Research Mentoring **Independent research presentations*

- 2025 Aaron Sides, Katja Treadwell
- 2024 Sean Cole*, Eli Oyler, Valerie Ng, Annika Peterson, Clare Reif, Jack Simon
- 2023 Tallulah Bowden*, Camille Gaston*, Luke Halpern, Ashley Jeong*, Carly Rauh, Philip Thomson, Sophia Trone
- 2022 Urooj Ahmed, Nouran Alim, Vanessa Amezcua, Jenna Antonacci*, Shelby Benz, Seamus Caslin*, Billie Maguire*, Cathilyn McIntosh*, Jean-Paul Nguyen, Mary Nguyen

Appointments

- 2025 – **Doctoral Research Assistant**, Department of Biology, Georgetown University
- 2022 – **Doctoral Teaching Assistant**, Department of Biology, Georgetown University
- 2021 **Adjunct Professor**, Environmental Science, Northern Virginia Community College
- 2021 **Field and Laboratory Research Assistant**, Department of Biology, Georgetown University
- 2020 – 21 **Graduate Teaching Assistant**, School of Environment and Natural Resources, Ohio State University

2019 – 20 **Patrick S. Osmer Fellow**, School of Environment and Natural Resources, Ohio State University
2018 **SROP Research Associate**, Big Ten Academic Alliance

Locational experience

United States: Delaware, District of Columbia, Maryland, Massachusetts, New Jersey, New York, North Carolina, Ohio, Rhode Island, Virginia, Washington

International: Norway, United Kingdom (England, Scotland)